

Curated closet: A Personalised Fashion Recommendation System for H&M using Machine Learning techniques.



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Abstract

This research focus on the development of a data drive personalised fashion recommendation system for H&M’s online retail platform. We use the provided transaction data, customer metadata and product information to create a machine learning model that helps us with improve the shopping experience for customers by providing a personalised recommendations. By analysis the user behaviour and product characteristics, the model aims to recommend relevant items, leading to customer satisfaction and increased sales.

Introduction

The rise of e-commerce platforms has both advantages and challenges for the fashion industry. H&M is working on developing a recommendation system that adapts to customer interests and preferences, making their shopping experience seamless. The research builds on previous projects in recommendation systems, using machine learning algorithms and innovative data processing techniques. By analyzing H&M's large data, which includes millions of client entries and product details, the study aims to find hidden patterns and insights for a successful recommendation system. The multidisciplinary approach, including data science, machine learning, and fashion analytics, aims to create an interesting, personalized, and seamless shopping experience in the digital world.

Aim

The aim of this research is to develop a dynamic machine learning model that effectively predicts the top 12 product recommendations for each customer over a 7-day period, leveraging both real-time and historical data. This model is designed to intuitively respond to the immediate needs of customers, ensuring a personalized and engaging shopping experience on H&M’s online platform.

Objective

To develop machine learning model that can predict the top 12 product recommendations for each customer within 7-days’ time interval. This model will use real-time and historic data to guess and fulfil the immediate needs of the customer to make sure an responsive and dynamic shopping experience.

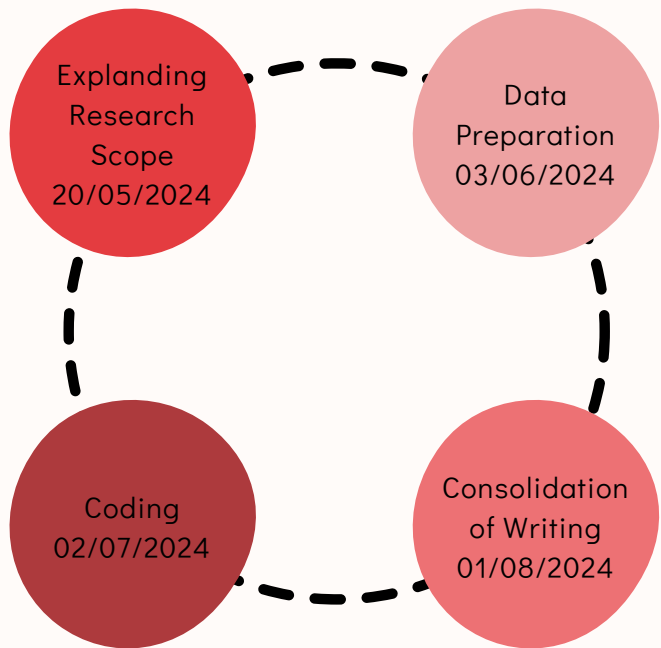
Research Question

How can we use machine learning to predict the top 12 product recommendations for each customer within 7-day time interval, resulting in a on-demand recommendation system for H&M?

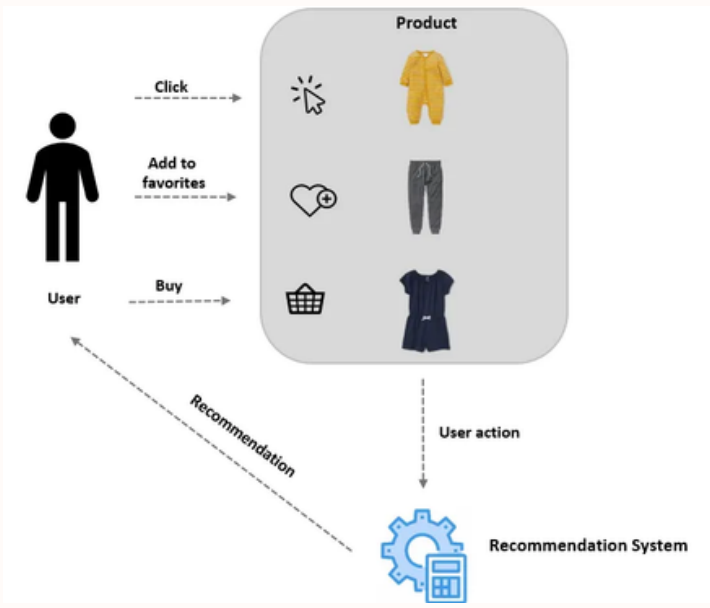
Hypothesis

Ho : There is no relationship between the machine learning- based recommendations and the actual products purchased by customers within 7-day time frame

H1: There is a major relationship between the machine learning- based recommendations and the actual products purchased by customer within 7- day time frame.



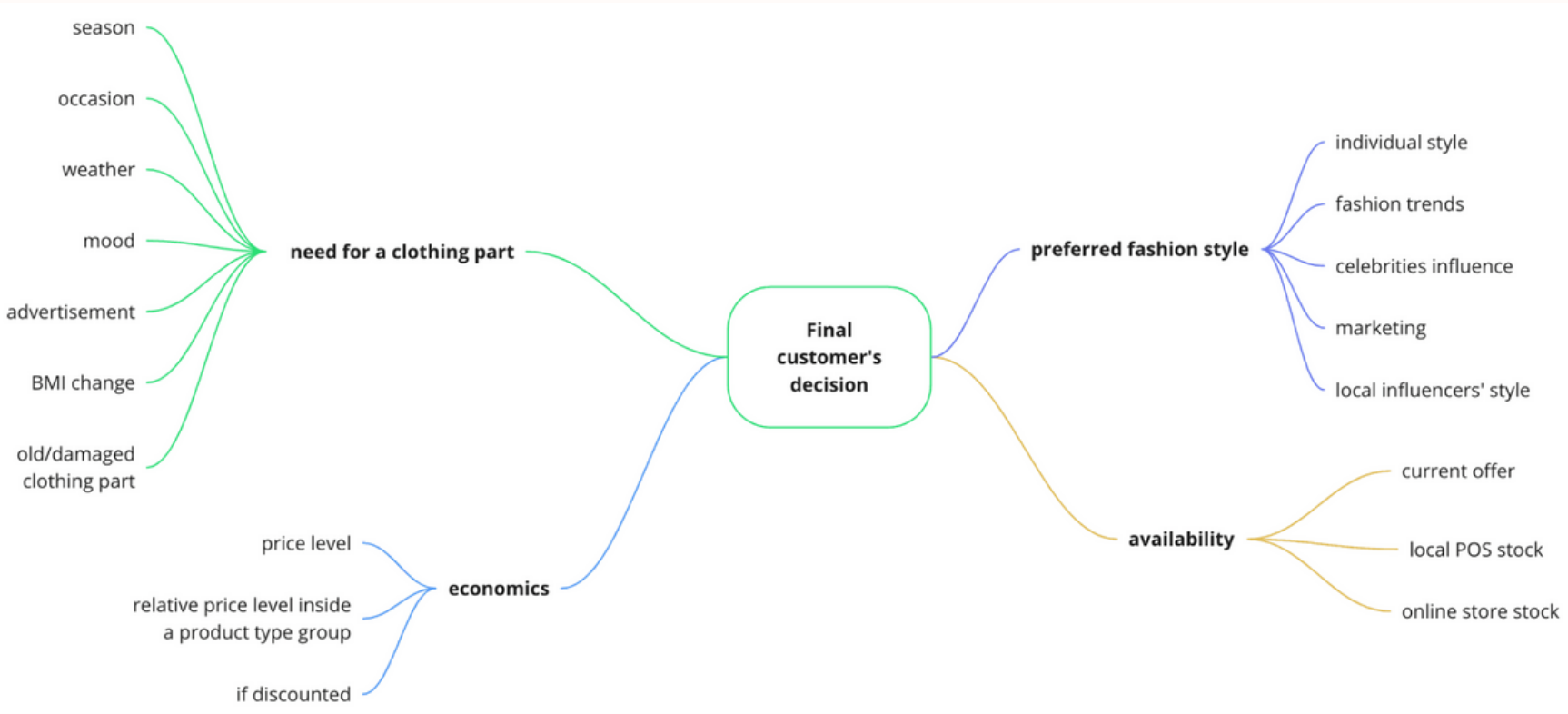
Timeline



Process of Recommendation System for fashion industry.

Methodology

This research aims to develop a personalized fashion recommendation system for H&M, predicting customer preferences and suggesting relevant items. The data sourced from H&M's Kaggle challenge is used for cross-sectional analysis and is explanatory and evaluative in nature. The system uses quantitative analysis techniques, including regression analysis, to identify factors influencing purchase behavior. Machine learning algorithms like LightGBM and CatBoost will be used to build predictive models, based on previous interactions with the platform. A hybrid approach, combining collaborative and content-based filtering, will be used to balance personalised user experiences with new item discovery, optimizing customer satisfaction and retention.



Mind map which shows factors that influence a customer’s decision

Conclusion

The development and implementation of a fashion recommendation system, using machine learning and data processing techniques, has the potential to transform the retail industry and improve customer shopping experiences. However, challenges such as changing fashion trends, visual aspects, and constant change are present. Despite these, the research emphasizes ethical considerations, transparency, and user empowerment in design and implementation. Retraining the model regularly is crucial to maintain accuracy and sales. The goal is to implement a model that can grow, gain customer loyalty, and provide a seamless shopping experience, ultimately increasing sales and providing a better shopping experience for consumers.



Summer Fashion recommendation

Reference

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